Semester project on control and simulation of automation systems

BEng in Robot Systems

4. Semester

Authors (Group 1?)

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Contents

1 Introduction

In life and industri that are alot of machinery that need regualted and precice controll, which i why a controller is made to achive each goal as needed. In robotics controllers can be expicaly usefull when trying to complete difficult taskes with multipul moveing parts. Where balance is one of main hindrences, when trying to design such a system. This projeckt will focus trying to solve a balance isuee of an inverted pendulem.

1.1 Project Goals

With a PLC, motor, conveyor track, inverted pendulem on a cart and a two diffrencet controller. it is attempted to make the inverted pendulem balance rightsideup using a controller to controll the force of the motor. success is mesurd by how well the controller balences the pendulem compared to a simulation of the controller, and the best controller is the one whose closest to the simulation of said controller.

1.2 Problem Definition

1.3 Constraints & Limitations

The limitations for this projeckt come from the physical setup, and the PLC. The cart is limited to move within two sensors on the conveyor track, and the maximum force for the moter is X. For the PLC it cant run faster than 1000Hz, and sampling via OPCUA is limited to 100Hz

1.4 System OverView

2 System modeling

2.1 Modelling of the Inverted Pendulum System

2.1.1 System Overview

The purpose of the modelling process is to obtain a mathematical representation of the inverted pendulum system, which can later be used for simulation and controller design. A mathematical model makes it possible to analyze the dynamic behavior of the system and evaluate different control strategies before implementation on the physical setup.

In this project, two classical modelling approaches were used: the Newton-Euler method and the Euler-Lagrange method. Both methods describe the dynamics of the system, but from different physical perspectives. The Newton-Euler method is based on force and moment balances, while the Euler-Lagrange method is based on the kinetic and potential energy of the system. The resulting nonlinear equations were linearized around the upright equilibrium position to simplify the controller design and implementation in MATLAB/Simulink.

The physical system consists of a cart with an attached inverted pendulum and an additional tip mass mounted at the end of the pendulum rod. The cart is constrained to move horizontally along a conveyor-driven rail, while the pendulum is free to rotate around its pivot point. The system is actuated by an external force F, which controls the horizontal movement of the cart. The goal of the system is to keep the pendulum balanced in its upright position while controlling the motion of the cart.

The generalized coordinates used to describe the system are:

- x: horizontal cart position
- θ: pendulum angle relative to the vertical axis

The physical parameters used in the model are shown in Table **??**.

The following assumptions were made during the modelling process:

- The pendulum rod is considered rigid
- Friction is assumed to be small and primarily represented by damping coefficients
- The pendulum motion is constrained to a two-dimensional plane
- The system is linearized around the upright equilibrium position

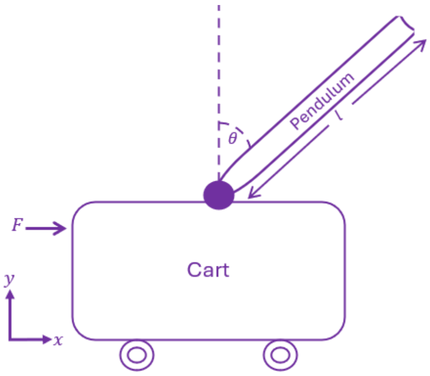

Figure 2.1: Simplified model of the inverted pendulum system showing the generalized coordinates, pendulum angle, and input force.

Table 2.1: System parameters used for modelling of the inverted pendulum system

Symbol	Parameter	Value	Unit
M	Mass of cart	0.500	kg
m	Mass of pendulum (including tip mass)	0.084	kg
l	Length of pendulum rod	0.35	m
L_{belt}	Length of conveyor belt	1.72	m
r	Radius of belt rollers	0.05	m
g	Gravitational acceleration	9.82	m/s^2
b_{lin}	Linear damping coefficient	5	N/(m/s)
b_{rot}	Rotational damping coefficient	0.0012	Nm/(rad/s)

2.1.2 Newton-Euler Modelling

The Newton-Euler method is based on applying Newton's second law and rotational moment equations to the system. Free body diagrams were constructed for both the cart and the pendulum to identify the forces and moments acting on the system.

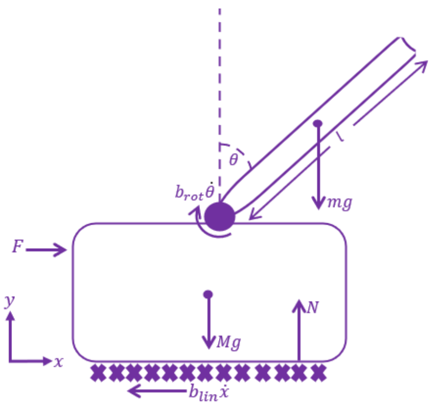

Figure 2.2: Free body diagram of the inverted pendulum system showing the applied force, gravitational forces, damping effects, and normal force acting on the system.

For the cart, the sum of forces in the horizontal direction was established using Newton's second law:

$$\sum F_x = M\ddot{x} \tag{2.1}$$

For the pendulum, both translational and rotational dynamics were considered. The pendulum experiences gravitational forces, reaction forces from the cart, and inertial effects caused by the cart motion.

To eliminate the internal reaction forces between the cart and pendulum, the moment equation was derived around the pendulum center of mass. This resulted in a coupled set of nonlinear differential equations describing the dynamic behavior of the system.

The obtained equations contain nonlinear trigonometric terms originating from the pendulum rotation. To simplify the controller design, the equations were linearized around the upright equilibrium position using the small-angle approximations:

$$\sin(\theta) \approx \theta \tag{2.2}$$

$$\cos(\theta) \approx 1 \tag{2.3}$$

After linearization, the system equations were transformed into transfer functions and state-space representations suitable for simulation and controller design in MATLAB/Python.

Final Linearized Model

$$(I + ml^2)\ddot{\theta} - mgl\theta = ml\ddot{x} \tag{2.4}$$

$$(M + m)\ddot{x} + b\dot{x} - ml\ddot{\theta} = u \tag{2.5}$$

State-Space Representation

$$\dot{x} = Ax + Bu \tag{2.6}$$

$$y = Cx + Du \tag{2.7}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-b(I+ml^2)}{(M+m)I+Mml^2} & \frac{m^2gl^2}{(M+m)I+Mml^2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{-mlb}{(M+m)I+Mml^2} & \frac{mgl(M+m)}{(M+m)I+Mml^2} & 0 \end{bmatrix} \tag{2.8}$$

$$B = \begin{bmatrix} 0 \\ \frac{I+ml^2}{(M+m)I+Mml^2} \\ 0 \\ \frac{ml}{(M+m)I+Mml^2} \end{bmatrix} \tag{2.9}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{2.10}$$

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.11}$$

2.1.3 Euler-Lagrange Modelling

The Euler-Lagrange method describes the system dynamics using energy relationships instead of direct force balances. The method is based on the kinetic and potential energy of the system.

The kinetic energy of the cart is given by:

$$T_{cart} = \frac{1}{2}M\dot{x}^2 \tag{2.12}$$

The pendulum contributes both translational and rotational kinetic energy due to its motion relative to the cart. Additionally, the pendulum possesses potential energy caused by gravity.

Using the derived expressions for kinetic and potential energy, the Lagrangian was formulated as:

$$L = T - V \tag{2.13}$$

The equations of motion were then derived using the Euler-Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i \tag{2.14}$$

where q_i represents the generalized coordinates of the system.

By applying the Euler-Lagrange formulation to both the cart position and pendulum angle, a coupled non-linear dynamic model of the system was obtained.

Similar to the Newton-Euler approach, the nonlinear equations were linearized around the upright equilibrium position to simplify further analysis and controller design.

Kinetic and Potential Energy

$$E_{kin} = E_{kin,c} + E_{kin,p} + E_{kin,rot} \tag{2.15}$$

$$= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\dot{x}^2 - ml\dot{\theta}\dot{x}\cos\theta + \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}I\dot{\theta}^2 \tag{2.16}$$

$$E_{pot} = mgl\cos\theta \tag{2.17}$$

Nonlinear Equations of Motion

$$-ml\ddot{x}\cos\theta + ml^2\ddot{\theta} + I\ddot{\theta} - mlg\sin\theta = -b_{rot}\dot{\theta} \tag{2.18}$$

$$(M+m)\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta = F - b_{lin}\dot{x} \tag{2.19}$$

Linearized Equations

$$ml^2\ddot{\theta} + I\ddot{\theta} - mlg\theta + b_{rot}\dot{\theta} = ml\ddot{x} \tag{2.20}$$

$$(M+m)\ddot{x} - ml\ddot{\theta} + b_{lin}\dot{x} = u \tag{2.21}$$

Transfer Function

$$\frac{\Theta(s)}{U(s)} = \frac{1}{\left((M+m)\cdot\frac{ml^2s^2+Is^2-mlg+b_{rot}s}{mls^2}\right)s^2 - mls^2 + b_{lin}\left(\frac{ml^2s^2+Is^2-mlg+b_{rot}s}{mls^2}\right)s} \tag{2.22}$$

2.1.4 Comparison of the Methods

Both modelling approaches resulted in equivalent dynamic models after linearization, verifying the correctness of the derivations.

The Newton-Euler method provided a more intuitive understanding of the physical forces and moments acting on the system, since it directly uses force and torque balances. However, the method becomes increasingly complex for systems with multiple interconnected bodies.

The Euler-Lagrange method provided a more systematic and compact mathematical formulation by describing the system through energy relationships. This makes the method particularly useful for more advanced multibody systems.

The final linearized model obtained from the derivations was used as the foundation for the controller design and simulations presented later in this project.

The final linearized model was implemented in MATLAB/Python and used for simulation and controller development. Both transfer function and state-space representations were used throughout the project for analysis and validation of the implemented controllers.

The derived model forms the basis for the PID controllers, pole-placement controller, and the discrete implementation used later in the project.

3 Control theroy

4 PLC implemetation

5 Data Collection

6 Evaluation

7 Discussion

8 Conclusion

References

A Distribution of Tasks

Table A.1: Distribution of Tasks

Task	**Noah**	**Rasmus**	**Emma**	**Jannick**	**Eymundur**
			Development		
System Modeling					
Control Theroy					
PLC implementation					
Data Collection					
			Report		
Abstract					
Introduction					
System Modeling					
Control theroy					
PLC Implementation					
Data Collection					
Results					
Discussion					
Conclusion					

B Code

The code is available in the attached ZIP file as well as on github at:

C Video Showcase

A video of the system running is available at: